THE $F_{\QCOH, *}$ ADJOINT AND ITS COHOMOLOGICAL $\DELTA$-FUNCTOR STRUCTURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites items from Homology plus five same-chapter results, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Right adjoint Q to the inclusion of $\QCoh$. Stacks Project, Cohomology of Algebraic Stacks, Lemma lemma-adjoint. Supplies the second and third adjunction steps and exactness of Q.
- Summary of LQCoh^{fbc} properties. Stacks Project, Cohomology of Algebraic Stacks, Proposition proposition-loc-qcoh-flat-base-change. Supplies membership of direct images and topology-independence.
- Exactness of quasi-coherent sequences up to parasitic modules. Stacks Project, Cohomology of Algebraic Stacks, Lemma lemma-exact-sequence-quasi-coherent-parasitic-cohomology. Justifies the four-way splitting.
- Pushforwards of parasitic modules are parasitic. Stacks Project, Cohomology of Algebraic Stacks, Lemma lemma-pushforward-parasitic. Kills the $R^i f_* \mathcal{P}_j$ terms.
- Kernel and cokernel of $Q(\mathcal{F}) \to \mathcal{F}$ are parasitic. Stacks Project, Cohomology of Algebraic Stacks, Lemma lemma-adjoint-kernel-parasitic. Explains why parasitic terms vanish under Q.
- Definition of cohomological δ-functor. Stacks Project, Homology, Definition homology-definition-cohomological-delta-functor. Supplies the target notion.

Source reference. Original proposition 0771: sources/stacks-chow/tags.txt, source lines 9186–9186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0778: sources/stacks-chow/tags.txt, source lines 9193–9193. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 010Q: sources/stacks-chow/tags.txt, source lines 1336–1336. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0776: sources/stacks-chow/tags.txt, source lines 9191–9191. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0775: sources/stacks-chow/tags.txt, source lines 9190–9190. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. The functor $f^* : QCoh(\mathcal{O}_\mathcal{Y}) \to QCoh(\mathcal{O}_\mathcal{X})$ has a right adjoint*

$$f_{QCoh,*} : QCoh(\mathcal{O}_\mathcal{X}) \longrightarrow QCoh(\mathcal{O}_\mathcal{Y})$$

which can be defined as the composition

$$QCoh(\mathcal{O}_\mathcal{X}) \to LQCoh^{fbc}(\mathcal{O}_\mathcal{X}) \xrightarrow{f_*} LQCoh^{fbc}(\mathcal{O}_\mathcal{Y}) \xrightarrow{Q} QCoh(\mathcal{O}_\mathcal{Y})$$

where the functors f_* and Q are as in Proposition 0771 and Lemma 0778. Moreover, if we define $R^i f_{QCoh,*}$ as the composition

$$QCoh(\mathcal{O}_\mathcal{X}) \to LQCoh^{fbc}(\mathcal{O}_\mathcal{X}) \xrightarrow{R^i f_*} LQCoh^{fbc}(\mathcal{O}_\mathcal{Y}) \xrightarrow{Q} QCoh(\mathcal{O}_\mathcal{Y})$$

then the sequence of functors $\{R^i f_{QCoh,*}\}_{i\geq 0}$ forms a cohomological δ-functor.

Proof. This is a combination of the results mentioned in the statement. The adjointness can be shown as follows: Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_\mathcal{X}$-module and let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_\mathcal{Y}$-module. Then we have

$$\begin{aligned} \mathrm{Mor}_{QCoh(\mathcal{O}_\mathcal{X})}(f^*\mathcal{G}, \mathcal{F}) &= \mathrm{Mor}_{LQCoh^{fbc}(\mathcal{O}_\mathcal{Y})}(\mathcal{G}, f_*\mathcal{F}) \\ &= \mathrm{Mor}_{QCoh(\mathcal{O}_\mathcal{Y})}(\mathcal{G}, Q(f_*\mathcal{F})) \\ &= \mathrm{Mor}_{QCoh(\mathcal{O}_\mathcal{Y})}(\mathcal{G}, f_{QCoh,*}\mathcal{F}) \end{aligned}$$

the first equality by adjointness of f_* and f^* (for arbitrary sheaves of modules). By Proposition 0771 we see that $f_*\mathcal{F}$ is an object of $LQCoh^{fbc}(\mathcal{O}_\mathcal{Y})$ (and can be computed in either the fppf or étale topology) and we obtain the second equality by Lemma 0778. The third equality is the definition of $f_{QCoh,*}$.

To see that $\{R^i f_{QCoh,*}\}_{i\geq 0}$ is a cohomological δ-functor as defined in Homology, Definition 010Q let

$$0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$$

be a short exact sequence of $QCoh(\mathcal{O}_\mathcal{X})$. This sequence may not be an exact sequence in $Mod(\mathcal{O}_\mathcal{X})$ but we know that it is up to parasitic modules, see Lemma 0776. Thus we may break up the sequence into short exact sequences

$$\begin{aligned} 0 \to \mathcal{P}_1 \to \mathcal{F}_1 \to \mathcal{I}_2 \to 0 \\ 0 \to \mathcal{I}_2 \to \mathcal{F}_2 \to \mathcal{Q}_2 \to 0 \\ 0 \to \mathcal{P}_2 \to \mathcal{Q}_2 \to \mathcal{I}_3 \to 0 \\ 0 \to \mathcal{I}_3 \to \mathcal{F}_3 \to \mathcal{P}_3 \to 0 \end{aligned}$$

of $Mod(\mathcal{O}_\mathcal{X})$ with $\mathcal{P}_i$ parasitic. Note that each of the sheaves $\mathcal{P}_j$, $\mathcal{I}_j$, $\mathcal{Q}_j$ is an object of $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$, see Proposition 0771. Applying $R^i f_*$ we obtain long exact sequences

$$\begin{aligned} 0 \to f_*\mathcal{P}_1 \to f_*\mathcal{F}_1 \to f_*\mathcal{I}_2 \to R^1 f_*\mathcal{P}_1 \to \ldots \\ 0 \to f_*\mathcal{I}_2 \to f_*\mathcal{F}_2 \to f_*\mathcal{Q}_2 \to R^1 f_*\mathcal{I}_2 \to \ldots \\ 0 \to f_*\mathcal{P}_2 \to f_*\mathcal{Q}_2 \to f_*\mathcal{I}_3 \to R^1 f_*\mathcal{P}_2 \to \ldots \\ 0 \to f_*\mathcal{I}_3 \to f_*\mathcal{F}_3 \to f_*\mathcal{P}_3 \to R^1 f_*\mathcal{I}_3 \to \ldots \end{aligned}$$

where are the terms are objects of $LQCoh^{fbc}(\mathcal{O}_\mathcal{Y})$ by Proposition 0771. By Lemma 0775 the sheaves $R^i f_* \mathcal{P}_j$ are parasitic, hence vanish on applying the functor Q, see Lemma 0779. Since Q is exact the maps

$$Q(R^i f_* \mathcal{F}_3) \cong Q(R^i f_* \mathcal{I}_3) \cong Q(R^i f_* \mathcal{Q}_2) \to Q(R^{i+1} f_* \mathcal{I}_2) \cong Q(R^{i+1} f_* \mathcal{F}_1)$$

can serve as the connecting map which turns the family of functors $\{R^i f_{QCoh,*}\}_{i \geq 0}$ into a cohomological δ-functor. $\square$